

Gaurav Totla

Lead Software Engineer

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LinkedIn

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To design and deliver scalable automation frameworks, secure back-end services, and reliable CI/CD pipelines by leveraging strong expertise in Python-based development, test automation, and DevOps practices within enterprise-grade environments.

Education

- 2021: Bachelor of Technology in Computer Science from Arya Institute of Engineering and Technology, Jaipur

Areas of Expertise

- Automation Framework Development
- Test Automation Strategy
- CI/CD Pipeline Optimization
- Back-end Service Development
- API Integration
- Infrastructure Provisioning
- Regression Testing
- Test Orchestration
- Environment Setup Automation
- Continuous Integration
- Quality Assurance Processes
- Release Validation

Technical Skills

- Programming Languages:** Python, JavaScript, C++, Shell Scripting, HTML, CSS
- DevOps & Cloud:** Jenkins, Docker, Ansible, AWS, Git, SVN
- Testing & QA Tools:** Robot Framework, Selenium, PyTest, Postman, JIRA, Playwright
- Frameworks & Libraries:** Flask, Django, Fast API
- Databases:** MySQL
- Operating Systems:** Linux
- Development Tools:** Visual Studio Code, PyCharm, GitLab

Awards

- Spot Award | Oct 2025
- Spot Award | Jun 2024

Personal Details

Date of Birth: 4th March 2000
Languages Known: English, Hindi
Address: Bangalore - 560064

Profile Summary

- Possess over 5 years of experience in building scalable automation solutions, back-end services, and CI/CD workflow across enterprise environments.
- Working as a **Lead Software Engineer at Collins Aerospace**, driving automation framework development, CI/CD optimization, and test infrastructure orchestration.
- Strong expertise in **Python-driven automation** and back-end development using **FastAPI**, along with **Robot Framework and Selenium** across distributed Linux systems.
- Proven capability in designing secure **RESTful APIs to streamline automation workflow and reduce manual validation effort**.
- Rich experience in **CI/CD pipeline architecture, parallel test execution**, and delivery optimization for continuous integration environments.
- Hands-on exposure to **infrastructure provisioning, configuration management**, and simulator-based testing aligned with enterprise compliance needs.

Work Experience

Collins Aerospace, Bangalore | 2021 - Present

Designation: Lead Software Engineer

Highlights:

- Engineered scalable Python automation frameworks using Robot Framework, Selenium, and Flask for reliable validation across distributed Linux environments.
- Deployed the application on AWS EC2 with Nginx as a reverse proxy, configuring Linux server, networking, permissions and backend process management.
- Designed a Python-based orchestration tool to execute, analyze, and consolidate 1,300+ test suites, cutting regression cycles from ~10 days to 4 days and accelerating delivery by 60%.
- Architected secure Flask REST APIs to enable automation workflow, improving integration efficiency and reducing manual validation effort by 50%.
- Implemented end-to-end CI/CD pipelines using Jenkins, Docker, and Git to automate unit, integration, and UI testing across environments.
- Optimized CI/CD pipelines through parallel execution and tuning, reducing total test runtime by 40% while improving pipeline stability.
- Automated 300+ manual test cases and maintained 800+ automated scenarios, increasing regression coverage beyond 90% and reducing production defect leakage.
- Authored Ansible playbooks to provision distributed Linux simulators, reducing environment setup time from 3-4 hours to under 10 minutes (95% faster).
- Used Gen AI/LLMs to generate test scenarios, edge cases, and test data, improving test coverage and reducing manual test-design effort.

Certifications

- AWS Certified Cloud Practitioner
- Complete DevOps Ansible Automation Training
- Linux Red Hat Certified System Administrator
- Docker for the Absolute Beginner - Hands-on - DevOps

Projects

News Blog | [Link](#)

- Developed a Django-based full stack news platform with secure authentication, content management, post creation, editing, dynamic commenting, and a scalable ORM-driven back-end.

URL Shortener | [Link](#)

- Developed a full stack URL shortener using React, Python, SQL Alchemy and SQLite, with Rest APIs for URL creation and deletion. Deployed the application on AWS EC2, configuring Nginx as a reverse proxy.